

Axle Components

10-4

Axle Angle Sensors

The axle angle sensors measure the axle angle and communicates that information to the ground controls ECM. There are two axle angle sensors. They are located on opposite axle pivot pins at each end of the chassis.

How to Install an Axle Angle Sensor

Note: If an axle angle sensor has been removed or replaced, the axle angle sensors will need to be calibrated.

Note: Perform this procedure with the axles fully retracted and the boom in the stowed position.

- 1 Locate the axle angle sensor to be replaced.
- 2 Tag and disconnect the angle sensor electrical connector from the chassis harness.
- 3 Remove the angle sensor cover retaining fasteners. Remove the angle sensor assembly.

Note: Inspect the sensor activator pin to make sure it is not broken or twisted.

Note: If the sensor activator pin needs to be replaced, install the new activator pin parallel with the drive chassis side plate. Refer to illustration 1.

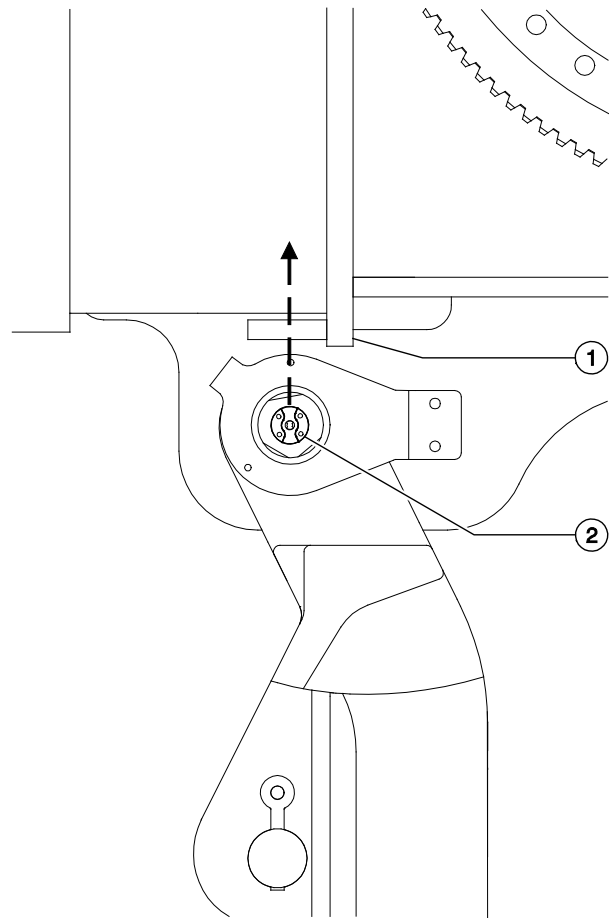


illustration 1

- 1 drive chassis side plate
- 2 activator pin

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- 4 Position the new angle sensor assembly over the sensor activator pin as shown. Refer to illustration 2.

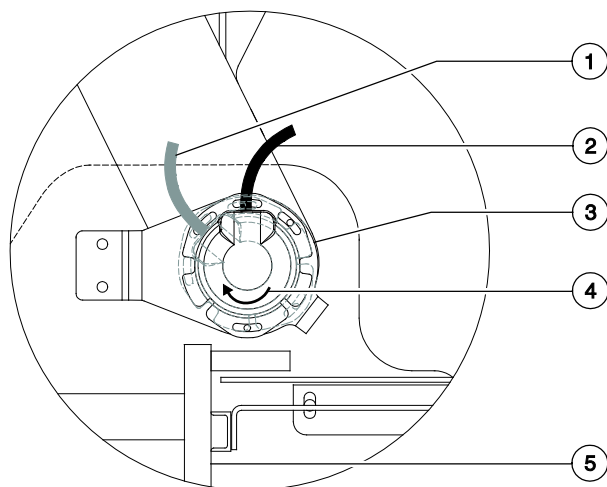


illustration 2

- 1 starting position
- 2 installed position
- 3 sensor cover
- 4 rotation arrow
- 5 drive chassis side plate

- 5 Align the sensor with the pin and install the sensor on to the pin.

Note: Be sure the actuator pin is engaged into the sensor.

- 6 Rotate the sensor housing in a clockwise direction until the sensor cable is parallel with the drive chassis side plate. Refer to illustration 2.
- 7 Install the axle sensor cover retaining fasteners. Do not tighten the retaining fasteners.
- 8 Connect the axle sensor assembly cable to the chassis harness.
- 9 Calibrate the axle angle sensor. Refer to Repair Procedure, *How to Calibrate the Axle Angle Sensors*.

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How to Calibrate the Axle Angle Sensors

Note: Use the following chart to identify the description of each LCD screen control button used in this procedure.



Plus



Minus



Previous



Enter

Note: If an axle angle sensor has been removed or replaced, the axle angle sensors will need to be calibrated.

Note: Perform this procedure with the axles fully retracted and the boom in the stowed position.

Note: Two people will be required to perform this procedure.

- 1 Turn the key switch to platform controls and pull out the red Emergency Stop button to the on position at both ground and platform controls.
- 2 Loosen the axle angle sensor cover retaining fasteners. Do not remove the fasteners or the sensor cover.
- 3 Using a voltmeter set to DC voltage, probe the back of the electrical connector at pins 2 and 3.

Note: If available, WebGPI can also be used for this procedure.

- 4 Rotate the sensor cover clockwise or counterclockwise until the voltage reading is between 4.2 to 4.4V DC. Tighten the sensor cover retaining fasteners.
- 5 Repeat steps 2 through 4 for the other axle angle sensor if needed.
- 6 Push in the ground controls red Emergency Stop button to the off position.
- 7 Open the ground control box.

Before serial number 321:

- 8 Locate the 2 pin connector in the lower left hand area of the ground control box.
- 9 Remove the connector plug and insert a jumper wire between the two pins of the connector.

After serial number 320:

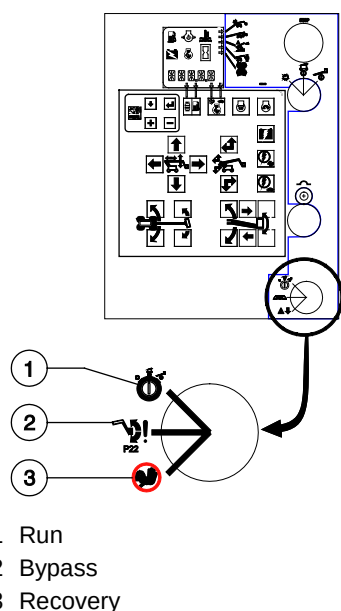
- 10 Locate the calibration toggle switch at the top of the ground control box. Activate calibration mode by moving the toggle switch in the left direction.
- 11 Temporarily install a control box door retaining fastener between the door and the box to prevent the control box door from moving the toggle switch while calibrating the machine.

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All models:

- 12 Remove the key from the main key switch. Insert the key into the bypass/recovery key switch and turn it to the bypass position.

Note: The angle sensor calibration values will not be saved correctly unless the key switch is in the bypass position and the calibration toggle switch is activated.



Note: The bypass/recovery key switch is located on the lower right side of the control box from serial number Z13512-1712.

- 13 Press and hold the **enter** button on the ground control panel while pulling out the ground controls red Emergency Stop button to the on position. Hold the **enter** button for approximately 5 seconds and then release it.
- 14 Enter sensor calibration mode by pressing the buttons at the ground controls in the following sequence: **(plus)(enter)(enter)(plus)**.
- 15 Press the **enter** or **previous** button on the LCD screen until DELETE AXLE ANGLE SENSORS CALIBRATION is displayed. Press the **plus** button to select YES, then press the **enter** button to accept.

Note: Models before software version 3.0, the LCD screen will display RESET AXLE ANGLE SENSORS.

- 16 At the AXLE ANGLES FULLY RETRACTED screen, press the **plus** button to select YES, then press the **enter** button to accept.
- 17 When the AXLE ANGLES FULLY EXTENDED screen is displayed, start the engine and fully extend the axles.

Note: If the system exits out of calibration mode when the engine is started, repeat step 13.

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- 18 Press the **plus** button to select YES, then press the **enter** button to accept.
- 19 Press the **enter** or **previous** button on the LCD screen until EXIT is displayed.
- 20 Press the **plus** button to select YES, then press the **enter** button to accept.
- 21 Press and hold the engine start button for approximately 5 seconds to shut off the engine and to save the calibration settings.

Note: Do not turn the engine off with the key switch or red Emergency Stop button or all calibration points or values will not be saved.

- 22 Wait approximately 20 seconds and turn the machine off by pressing the red Emergency Stop button in.

Before serial number 321:

- 23 Remove the wire jumper from the 2 pin connector in the lower left hand area of the ground control box that was installed in step 8. Install the connector plug.

DANGER

Tip-over hazard. Failure to remove the jumper wire will result in an unsafe configuration which will compromise machine stability. Operating the machine in this configuration will result in the machine tipping over, resulting in death or serious injury. The jumper wire must be removed before operating the machine.

- 24 Close the ground control box and install the door retaining fasteners.

After serial number 320:

- 25 Remove the fastener that was temporarily installed. Close the control box door and install the door retaining fasteners.

Note: When the control box door is closed, the calibration toggle switch is automatically activated to exit out of calibration mode.

All models:

- 26 Turn the key back to the run position and remove the key from the bypass/recovery key switch. Insert the key into the main key switch and turn it to ground controls.

Note: Be sure that the bypass/recovery key switch is in the run position before attempting to operate the machine.